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Optimizing satellite-based cropland area estimation through integrated map accuracy assessment and stratified sampling design across six African countries

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ABSTRACT

Accurate cropland area estimation is essential for food security, yet conventional surveys are costly. Satellite-derived area estimates offer a scalable alternative; however, "pixel-counting" from satellite products introduces bias, while probability-based sampling with design-based inference provides unbiased area estimates, but its efficiency depends on map quality. In this study, we introduce an integrated framework linking map accuracy with relative efficiency to optimize stratified sampling designs. We evaluated seven land cover products across six African countries using independent reference data. Our results demonstrate that map selection is a critical determinant of survey costs: the most efficient products (predominantly GLAD and Digital Earth Africa) reduced required sample sizes by 20%–40% compared to less efficient alternatives while maintaining a target 10% coefficient of variation. We found that for rare cropland classes, gains in producers' accuracy improved sampling efficiency more than equivalent gains in users' accuracy. Our sample-based estimates ranged from 1.41 Mha (Rwanda) to 12.66 Mha (Tanzania) and correlated strongly with FAOSTAT arable land statistics (R^2 = 0.89). Our framework minimizes the sample size required to achieve a target precision, providing an operational-ready, cost-effective, and statistically rigorous guide for national agricultural monitoring in resource- and data-limited regions.

1. Introduction

Accurate estimation of cropland areas is critical for agricultural monitoring, food security assessment, economic planning, and environmental sustainability. Conventionally, cropland area estimates are derived from agricultural surveys and census data (Potapov et al., 2022; Tubiello et al., 2021; Carfagna et al., 2013). While essential for providing detailed and accurate information about agricultural practices and production, conventional approaches are expensive to conduct, require months of fieldwork, and demand significant human and financial resources (Hu et al., 2021; d'Andrimont et al., 2020; Gallego, 2012). In recent decades, satellite-derived datasets have emerged as a complementary source of information on cropland areas (Potapov et al., 2022; Lobell et al., 2020). Satellite imagery provides synoptic, timely, and cost-effective observations of the Earth's surface, enabling the mapping and monitoring of croplands at various spatial and temporal scales (H. Kerner et al., 2024; Potapov et al., 2022; Nakalembe et al., 2021). This is especially critical in regions where field-based surveys are inconsistent and underfunded (Kebede et al., 2024).

Several global and regional cropland maps have been developed in recent years using satellite data, including the Global Food Security-Support Analysis Data (GFSAD) (Oliphant et al., 2022), GLAD (Potapov et al., 2022), WorldCereal (Van Tricht et al., 2023), and the Esri Land Cover product (Esri and Impact Observatory, 2024). Region- and country-level efforts include Digital Earth Africa's cropland map (Burton et al., 2022), a binary cropland map for Togo (Kerner et al., 2020), and a Kenya map (Tseng et al., 2020). These products offer new opportunities for estimating cropland areas across large regions. Direct area estimation from satellite-based maps using pixel-counting, however, is biased due to classification errors (commission and omission), spatial resolution limitations, temporal observation periods, spectral properties, and cloud coverage (Whitcraft et al., 2015; Olofsson et al., 2014;

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Carfagna et al., 2013; McRoberts, 2011). The remote sensing community has developed statistically sound methods based on probability sampling to assess map accuracy and estimate area while communicating uncertainties using unbiased estimators (CEOS WGCV, 2025; Stehman and Foody, 2019; Olofsson et al., 2014; Stehman, 2014; Cochran, 2002-01-03). In particular, stratified random sampling has been recommended for improving the precision of sample-based area estimates, especially when rare classes (e.g., cropland in regions where cropland proportion is small relative to non-cropland) are used for stratification (Stehman, 2014; Olofsson et al., 2014).

A fundamental advantage of stratified random sampling is its efficiency: by using map-based stratification, researchers can achieve the same level of precision with a smaller sample size compared to simple random sampling or less efficient stratification approaches (Skakun, 2025; Olofsson et al., 2014). This efficiency translates directly into reduced costs, as collecting and labeling reference data, whether through field visits, high-resolution imagery interpretation, or expert annotation, is time-consuming, labor-intensive, and expensive (Stehman and Wagner, 2024; Gallego et al., 2014). The practical implication is clear: a more efficient map reduces the number of reference samples needed to reach a target precision, thereby lowering operational costs while maintaining statistical rigor.

In the Sub-Saharan African context, where agricultural landscapes are characterized by sparse and fragmented cropland dominated by smallholder farming (Giller et al., 2021; Fritz et al., 2015), cropland area estimation efforts using satellite-based approaches have reported high coefficients of variation (27%–42%), reflecting substantial imprecision (Tubiello et al., 2023b). Traditional statistical methods remain prohibitively expensive in these settings (McRoberts and Tomppo, 2007), with monitoring costs varying substantially based on precision requirements (Böttcher et al., 2009). This level of uncertainty limits the operational utility of area estimates for decision-making and reporting. Moreover, when sampling designs are not optimized to meet specific objectives, the result is either overly costly fieldwork or low precision.

A key operational question remains: what makes one land-cover map better than another for stratification, especially when multiple global and regional products are available with different reported performance metrics (H. Kerner et al., 2024)? Specifically, how do standard classification accuracy metrics, such as user's accuracy (UA) and producer's accuracy (PA), affect a map's efficiency for sample-based area estimation? Skakun (2025) demonstrated that the relative efficiency of stratification depends nonlinearly on both PA and UA, but that their contributions differ depending on the target class proportion. For minor or rare classes (e.g., cropland occupying a small fraction of the landscape), improvements in PA have a greater impact on efficiency than improvements in UA. Conversely, for major classes, UA becomes more influential. Importantly, increasing UA at the expense of PA can actually reduce a map's efficiency for area estimation (Skakun, 2025). These findings indicate that high classification accuracy alone does not guarantee better sampling outcomes unless gains in accuracy align with estimation objectives (Skakun, 2025; Stehman and Wagner, 2024; Gallego et al., 2014).

Building on these theoretical insights and recommended best practices (Stehman and Foody, 2019; Olofsson et al., 2014; Stehman, 2014), this study presents an integrated framework that combines map accuracy assessment, relative efficiency analysis, and stratified area estimation to improve the precision and cost-effectiveness of cropland area estimates. We apply this framework across six African countries characterized by smallholder agriculture and limited official cropland area statistics. Our methodology proceeds in three stages: (1) we assess the accuracy of multiple publicly available land cover maps using an independent stratified reference sample; (2) we quantify the relative efficiency of each map for stratification based on its PA and UA values, following Skakun (2025); and (3) we select the most efficient map for each country, use it as the basis for stratified random sampling, and estimate cropland area with a target coefficient of variation of 10%. By identifying which maps offer the greatest efficiency gains – thereby minimizing sample size requirements and associated costs – we demonstrate how strategic map selection can reduce resource demands while maintaining rigorous area estimates. Finally, we compare our sample-based estimates with national and international agricultural statistics to contextualize the results, though this comparison is exploratory rather than central to our methodological contribution.

2. Study area and datasets

2.1. Study area

We focus on six countries in Eastern and Southern Africa: Kenya, Malawi, Rwanda, Tanzania, Uganda, and Zambia (Fig. 1). These countries are part of the Regional Food Balance Sheet (RFBS) initiative, which provides seasonal crop production forecasts to inform agricultural policy decisions (Agra and Comesa, 2022).

Eastern and Southern Africa encompasses diverse agricultural landscapes shaped by contrasting climatic patterns. Kenya, Rwanda, Uganda, and northern Tanzania are characterized by bimodal rainfall regimes, with long rains from March to May and short rains from October to December. While Southern Tanzania, Malawi, and Zambia experience unimodal rainfall patterns, with a single rainy season extending from November to April (Palmer et al., 2023). These climatic variations drive distinct farming practices across the region, which is dominated by smallholder agricultural systems with average farm sizes from under one hectare to less than 10 hectares (Giller et al., 2021; Fritz et al., 2015). Agricultural land comprises approximately 12% of the total land area in Eastern Africa and 6% in Southern Africa (Tubiello et al., 2023a; Nzabarinda et al., 2021).

2.2. Cropland data

We analyzed open-access global and regional cropland data, as well as cropland layers from land cover and land use (LCLU) maps, for the year 2019. We selected 2019 as our reference year because it coincided with the concurrent availability of multiple high-quality products, enabling a comprehensive comparative assessment across diverse mapping efforts within a unified framework. The cropland datasets we analyzed included: Digital Earth Africa (10 m resolution, continental coverage) (Burton et al., 2022), and GLAD (30 m, global) (Potapov et al., 2022) cropland, and the cropland class from the following LCLU datasets; DynamicWorld (10 m, global) (Brown et al., 2022), Esri-LULC (10 m, global) (Esri and Impact Observatory, 2024), Copernicus (100 m, global) (Buchhorn et al., 2020), and GLC_FCS30D (30 m, global (Zhang et al., 2024)). Each LCLU map employs a distinct cropland definition ranging from annual herbaceous crops to broader agricultural land categories summarized in Table B.1. Detailed descriptions of the underlying models, methodologies, and classification definitions are included in H. Kerner et al. (2024). We converted each dataset into binary cropland/non-cropland classifications and clipped them to national boundaries, then generated a majority vote ensemble map for each country. We summed the binary cropland predictions across all six maps for each pixel and classified a pixel as cropland only if a strict majority of maps (i.e., >50%; at least 4 out of 6) predicted cropland. Ties (3 cropland vs. 3 non-cropland) were thus resolved in favor of non-cropland. The pixel values at reference sample locations were extracted using Google Earth Engine's reduceRegions() function, maintaining each map's native spatial resolution.

2.3. Agricultural land use data

Table 1 presents aggregated national data for our target countries. We compare our sample-based cropland area estimates to established

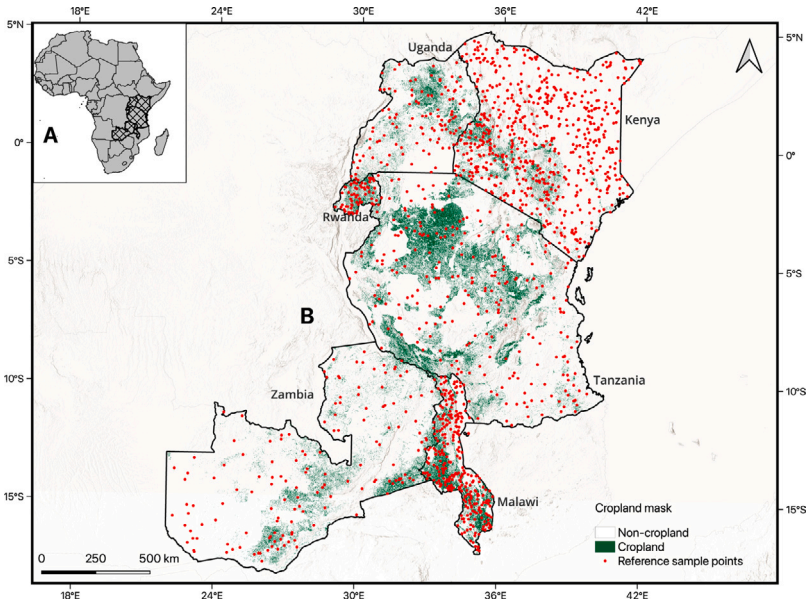


Fig. 1. Study area and spatial distribution of reference samples. **A** highlights the six focus countries in Africa: Kenya, Rwanda, Malawi, Tanzania, Uganda, and Zambia. **B** shows detailed independent binary cropland/non-cropland maps at 10 *m* resolution for each country, with green pixels representing predicted cropland. Reference sample locations are overlaid to indicate spatial coverage used for sample-based area estimation.

Table 1
Cropland area estimates from multiple data sources for 2019. Values are shown for FAOSTAT, USDA FAS, and official agricultural statistics, all in million hectares (Mha).

Country	FAOSTAT		USDA FAS	Official report
	Cropland	Arable land		
Kenya	6.41	5.8	2.78	6.36
Rwanda	1.5	1.19	0.47	1.11
Malawi	4.08	3.88	2.22	N/A
Tanzania	15.31	13.32	6.96	16.72
Uganda	9.1	6.9	2.39	4.66
Zambia	3.84	3.8	1.37	2.67

Note: USDA FAS figures represent the annual harvested area of major commodity crops only. "N/A" indicates data not available.

agricultural land use statistics from three primary sources: FAO's FAOSTAT database, the United States Department of Agriculture Foreign Agricultural Service (USDA FAS), and official national statistics from each country's agricultural or statistics department (where publicly available).

We acknowledge that agricultural statistics themselves contain uncertainties stemming from varying survey methodologies, sampling designs, and reporting practices across countries. The reliability of international agricultural databases depends on the quality of national data systems (FAO, 2025; Tubiello et al., 2023a), which can suffer from inconsistencies in reporting methods and statistical procedures (APEC, 2013), as well as misreporting and varying data availability (Cao et al., 2025; FAO, 2023).

2.3.1. *Official national agricultural statistics*

We obtained cropland area data from national statistical agencies and agricultural ministries, drawing from coordinated agricultural censuses and sample surveys conducted by each country's authorities for Kenya (Ministry of Agriculture & Livestock Development, 2019), Rwanda (National Institute of Statistics of Rwanda, 2019), Tanzania (National Bureau of Statistics, Tanzania, 2021), Uganda (Uganda Bureau of Statistics, 2020), and Zambia (Zambia Statistics Agency, 2020). The reported cropland statistics exhibit considerable variation in definition and aggregation across countries, with areas reported as planted area, crop cultivated area, or utilized agricultural area, typically aggregated by crop type or sub-administrative divisions (provinces or counties) (Table A.1). This variation reflects differences in national agricultural monitoring systems and reporting conventions. Despite extensive searches of national statistical databases and ministry publications, comparable official statistics for Malawi's 2019 cropland area were unavailable.

2.3.2. *FAOSTAT*

We retrieved 2019 cropland and arable land data for each study country from FAOSTAT (https://www.fao.org/faostat/), a free-access database of global food and agriculture statistics spanning 1961 to 2025. FAOSTAT's land use categories follow internationally standardized definitions (FAO, 2023): *cropland* encompasses all land used for cultivation of crops, including both *arable land* (temporary crops, temporary meadows and pastures, and temporary fallow) and *permanent crops* (such as coffee, tea, and oil palm plantations). *Arable land* thus represents a subset of cropland that excludes woody perennial crops but includes temporary pastures within crop rotations.

We included both cropland and arable land categories for three reasons. First, Tubiello et al. (2023b) demonstrated that definitional alignment significantly impacts comparison outcomes, with many remote sensing products showing closer correspondence to arable land than to the broader cropland category. Second, our reference data definition (Section 2.4) includes both temporary crops and plantations, theoretically aligning with FAOSTAT's cropland rather than arable land definition. Third, Africa exhibits substantial data limitations in FAOSTAT's land use domain, with only an 11% reporting rate and 84% imputation rate for missing data (FAO, 2023). Including both categories, therefore, provides more comprehensive coverage for assessing definitional impacts on agreement patterns. While the IPCC rates FAOSTAT as reasonably accurate, it encourages cross-validation against country-specific data sources (Galbally, 2000), which supports our approach of comparing multiple statistical sources.

2.3.3. *USDA FAS*

The USDA FAS International Production and Assessment Division (IPAD) assesses the condition, production, area estimates, and yield of

specific commodity crops grown worldwide. These assessments integrate multiple sources of information, including official country and news reports, satellite imagery, weather data, and derived models (USDA FAS, https://ipad.fas.usda.gov).

USDA FAS provides information on the annual harvested area of major commodity crops in individual countries at both national and sub-national levels. In our analysis, the USDA FAS cropland area data represent the total annual harvested area of all major commodity crops listed for each country. Although this dataset does not include other arable crops grown in our focused countries, it serves as a valuable indicator and source for cropland areas for research and agricultural market analysis (Franch et al., 2022).

2.4. Reference sample labeling

For each country's reference sample datasets, at least two members of our team independently annotated each sampling unit using multi-temporal high-resolution satellite imagery, including PlanetScope monthly composites (3 m spatial resolution), Google Earth Imagery, and other auxiliary sources available on Collect Earth Online.

We labeled active cultivated land within 12 months of each country's crop calendar as cropland. Following H.R. Kerner et al. (2024), this included identifying land that was sown/planted and harvestable at least once within 12 months. Critically, we also included plantations (such as coffee, tea, and oil palm) where these represented actively managed agricultural land, thereby aligning our operational definition with FAOSTAT's broader *cropland* category (which encompasses both temporary crops and permanent crops) rather than only with *arable land* (FAO, 2023; Tubiello et al., 2023a).

This definitional choice has important implications for comparing our results with different land cover products. Following Tubiello et al. (2023b), we recognize that products vary substantially in their treatment of permanent crops: some exclude them entirely (e.g., GLAD), while others include them with varying levels of detail (e.g., GLC-FCS30). Our definition most closely aligns with products that include permanent crops, though the relatively low prevalence of plantations in most of our study countries limits the practical impact of this definitional difference on most accuracy metrics.

When annotators disagreed on classifications, we conducted further reviews to reach consensus, utilizing all available evidence (H.R. Kerner et al., 2024; Finegold and Ortmann, 2016).

3. Methods

Our framework, summarized in Fig. 2, combines established best practices in map accuracy assessment and sample-based area estimation to generate reliable estimates of cropland area for our focus countries. Our approach consists of two main phases, each using a separate, independent reference dataset. First, we evaluate existing cropland maps using stratified estimators to assess the accuracy of cropland classification. Second, we design a stratified random sampling scheme that leverages map-based efficiency to estimate national cropland area with quantified uncertainties. From sample design and reference data interpretation to accuracy assessment and final area estimation, we employ statistically unbiased estimators to ensure both precision and reliability in quantifying cropland extent. Two independent samples were used—one for the initial map intercomparison (accuracy assessment) and a second for the final area estimation.

3.1. Cropland map accuracy assessment

To assess the accuracy of cropland maps for our focus countries, we implemented a stratified random sampling design for reference data collection to evaluate 2019 cropland datasets (Table B.1). Stratified random sampling is used to ensure a relatively large sample size in the rare class, thereby achieving a smaller standard error for accuracy estimates. Skakun (2025), Stehman and Foody (2019), Olofsson et al. (2014). The rare class in this study refers to cropland, which typically constitutes $< 35\%$ of total land area in five of the six study countries.

We stratified the study area using independent binary cropland maps for each country created using a multi-headed Long Short-Term Memory (LSTM) in-season time-series classification model (Tseng et al., 2020). From these defined strata, we selected sample pixels using stratified random sampling. This independent stratification map serves two key purposes: it provides an efficient sampling framework by concentrating samples in areas likely to contain cropland, and it establishes a common reference framework for evaluating multiple existing cropland maps. Since the stratification of our study area uses a map that differs from those we evaluate, the strata do not correspond exactly to the map classes of the evaluated maps. Consequently, we cannot use conventional stratified estimators that assume the stratification map and evaluated maps are the same (Tsendbazar et al., 2021; Stehman and Foody, 2019; Stehman, 2014). Instead, we apply the stratified estimators of Stehman (2014) when the strata differ from the map classes. Using these estimators, we calculate user accuracy (UA, precision), producer accuracy (PA, recall), overall accuracy (OA), F1 score (harmonic mean of precision and recall), and their associated standard errors for all evaluated land-cover maps, including ensemble products. Complete implementation details are provided in the supplementary repository (Adebayo et al., 2025).

We estimated reference sample size for each country using Eqs. (1) and (2) (Olofsson et al., 2014, Eq. (13)):

$$n = \left(\frac{\sum w_i \times s_i}{SE(\hat{O})} \right)^2, \tag{1}$$

$$s_i = \sqrt{u_i(1 - u_i)}. \tag{2}$$

We obtain both the weight of strata w_i (i.e., the proportion of area mapped as stratum i) and the stratum user accuracy u_i from the independent binary cropland map, where s_i is the standard deviation of stratum i. The $SE(\hat{O})$ is the desired standard error of the estimated overall accuracy (the suggested value is 0.01) (Olofsson et al., 2014).

Table 2 summarizes the reference datasets generated for accuracy assessment, including key construction parameters: stratum weights and user accuracy values from the independent binary cropland maps used for stratification. For sample allocation across strata, we adopted a compromise approach following Olofsson et al. (2014) rather than using standard proportional or equal allocation methods. Proportional allocation generally produces smaller standard errors for PA and overall accuracy than equal allocation, but it provides poor estimates for rare classes, whereas equal allocation favors UA estimates but can be inefficient. Our approach increases sample allocation to the cropland stratum relative to proportional allocation while keeping it below equal allocation levels, thereby expected to maintain small standard errors across all accuracy metrics (Olofsson et al., 2014). Using this allocation strategy, we generated a stratified random sample of 10×10-m pixels (our spatial assessment unit) from each country's binary cropland map classes. These initial reference samples used for map evaluation are statistically independent from the second set of samples collected for final area estimation (see Section 3.3). Within each stratum, sample pixels were selected using simple random sampling without replacement. We used these reference samples exclusively for the accuracy assessment of existing cropland maps and are distinct from those used for area estimation (Section 3.3). The resulting accuracy metrics (PA and UA) and cropland proportions from these reference samples provide input for evaluating map efficiency in area estimation, as discussed in Section 3.2.

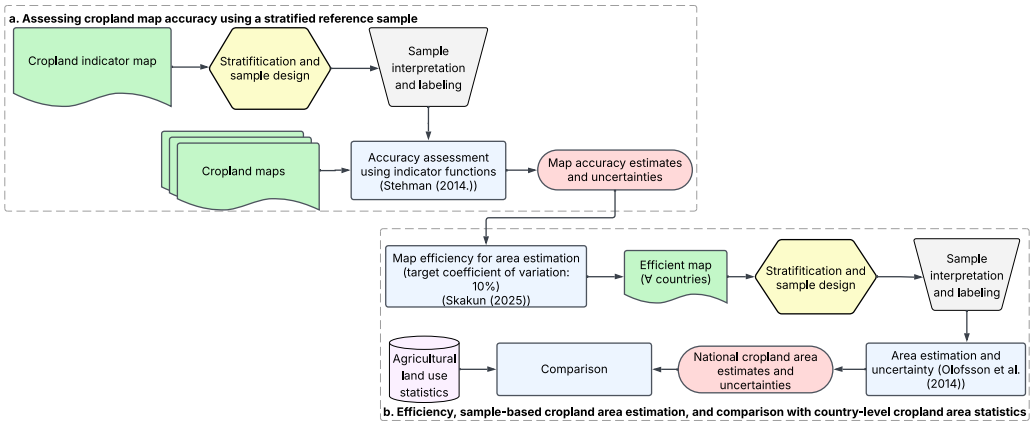


Fig. 2. Overview of the methodological workflow from **a** assessing the accuracy of existing cropland maps using a stratified reference sample, **b** evaluating efficiency, sample-based area estimation, and comparison with country-level cropland area estimates.

Table 2
Reference sample datasets for independent evaluation of cropland maps stratified using independent binary cropland maps, with each sampling unit (10 × 10-m pixel) validated through visual interpretation of high-resolution satellite imagery following regional crop calendars. Sample allocation parameters: (w_i) map proportion, (U_i) UA calculated from the independent binary map.

Country	Strata	weight (w_i)	U_i	Sample size	Label validity period
Kenya	Crop	8%	0.464	139	Feb 2019–Jan 2020
	Non-crop	92%	0.946	405	
Rwanda	Crop	46%	0.683	254	Jan 2019–Dec 2019
	Non-crop	54%	0.542	271	
Malawi	Crop	37%	0.461	134	Sep 2019–Aug 2020
	Non-crop	63%	0.937	376	
Tanzania	Crop	21%	0.574	184	Jan 2019–Dec 2019
	Non-crop	79%	0.882	412	
Uganda	Crop	23%	0.495	213	Jan 2019–Dec 2019
	Non-crop	77%	0.727	412	
Zambia	Crop	7%	0.765	214	Jan 2019–Dec 2019
	Non-crop	93%	0.779	346	

3.2. Evaluation of map efficiency for area estimation

We adopt the framework of Skakun (2025) to quantitatively assess how classification accuracy (map quality) impacts the efficiency of stratification for area estimation. Our objective is to minimize the standard error of cropland area estimates such that the coefficient of variation (CV) does not exceed 10%. The choice of target CV represents an operational and policy decision that balances precision requirements with resource constraints. While users may target a more stringent 5% CV for higher precision, this would require larger sample sizes depending on the proportion of the stratum (f) to be collected or labeled for the estimates. For this study, we adopt the 10% CV, which provides a reasonable balance between estimation precision and sampling feasibility.

Following Skakun (2025, Eq. 26), we define the target variance of the estimator as:

$$V_{\text{target}} = (\gamma f)^2, \tag{3}$$

where γ is the target coefficient of variation (10%) and f is the true proportion of the cropland class. In this study, we derived f from the area proportion of reference samples collected in Section 3.1 for each country. While we use reference-sample-based estimates, f can also be obtained from prior knowledge, such as published estimates from previous studies, pilot samples, or official agricultural statistics.

We evaluate the relative efficiency ($\eta^k_{A,B}$) of Map A compared to Map B as the ratio of sample sizes required to achieve identical precision in area estimation for class k:

$$\eta^k_{A,B} = \frac{n^B_{\text{opt}}}{n^A_{\text{opt}}}, \tag{4}$$

where n_{opt} is the optimal total sample size needed to meet the target variance V_{target}, defined as (Skakun, 2025):

$$n_{\text{opt}} = \frac{\left(\sum_{i=1}^{K} w_i s_i\right)^2}{V_{\text{target}}} = \frac{\left(\sum_{i=1}^{K} w_i \sqrt{u_i(1-u_i)}\right)^2}{V_{\text{target}}}, \tag{5}$$

where w_i is the proportion of total mapped area in stratum i, u_i is the user accuracy for stratum i, and $K = 2$ represents a binary stratification scheme. Specifically, $i = 1$ corresponds to the cropland stratum (the target class of interest) and $i = 0$ corresponds to the non-cropland stratum (all other land cover types aggregated). We derived these values from individual map evaluations described in Section 3.1.

A relative efficiency value $\eta^k_{A,B} > 1$ indicates that Map A requires a smaller sample size than Map B to achieve equivalent precision, translating to reduced costs in time, labor, and financial resources for reference data collection (Skakun, 2025). Using the n_{opt} of the most efficient map, we apply Neyman allocation to determine stratum-specific sample sizes, assuming equal sampling costs across strata (Skakun, 2025; Olofsson et al., 2014; Cochran, 2002-01-03).

3.3. Cropland area estimation

For each country, we collected a second, independent set of reference samples specifically for cropland area estimation. We selected the most efficient map identified in Section 3.2 to serve as the basis for stratification. This map's estimated optimal total sample size and strata allocation (required to achieve our target variance of a 10% coefficient of variation) determined our sampling design. We emphasize that this second reference sample is statistically independent from the first sample used for accuracy assessment (Section 3.1). Using stratified random sampling with the optimal map as the stratification basis, we selected and manually labeled a new set of reference pixels. We calculate the population error matrix to estimate the cropland area and associated uncertainty (±1 standard error), following Olofsson et al. (2014, Eq. 10 and 11) based on our manually labeled reference dataset described in Section 2.4.

We estimate the area of the cropland class ($\hat{A}_k$) according to Olofsson et al. (2014) using

$$\hat{A}_k = A_{tot} \times \sum_{i=1}^{2} w_i \frac{n_{ik}}{n_{i\cdot}}, \tag{6}$$

where A_{tot} is the total map area, w_i is the proportion of the mapped area in stratum i, n_{ik} is the number of reference samples in stratum i classified as class k based on the reference data, and $n_{i\cdot}$ is the total number of reference samples in stratum i. And the standard error (uncertainty) of the area estimate for cropland (class k) is estimated by Olofsson et al. (2014, Eq. 10):

$$SE(\hat{A}_k) = A_{tot} \times \sqrt{\sum_{i=1}^{2} w_i^2 \times \frac{\frac{n_{ik}}{n_{i\cdot}}\left(1 - \frac{n_{ik}}{n_{i\cdot}}\right)}{n_{i\cdot} - 1}}. \quad (7)$$

For the quantitative evaluation and comparison between sample-based cropland area estimates and agricultural land use statistics (Section 2.3), we compared our sample-based estimates against four different statistical sources: FAOSTAT Cropland, FAOSTAT Arable land, country official reports, and USDA-FAS. Using data from all countries in our study, we calculated and reported the coefficient of determination (R^2) and the normalized root mean square error (NRMSE, expressed as a percentage and computed as the RMSE divided by the RMSE range).

4. Results

4.1. Map accuracy assessment

In Table 3, we present comprehensive accuracy assessment results for the cropland maps we evaluated using reference samples from 2019 (detailed in Section 2.4). The final row provides summary statistics across all countries. We also evaluate an ensemble map generated by majority voting among all individual map predictions (Section 3.1).

To identify the best-performing maps, we ranked each map across all metrics and countries, highlighting the top-scoring maps (bold blue) and their second-highest scores (bold black) for each metric-country combination. The GLAD cropland dataset demonstrated the most consistent superior performance across the study region. Analyzing the individual metric-country combinations, GLAD achieved 17 first- or second-place rankings, with 9 first-place (bold blue) and 8 second-place (bold black) finishes. Digital Earth Africa emerged as the second-best performer with 15 total rankings (6 first-place and 9 second-place), showing consistently strong results across multiple contexts. Both the Esri LULC map (13 total rankings) and the ensemble map (11 total rankings) also performed competitively. In contrast, Dynamic World showed more variable performance, generally scoring lower in PA, OA, and F1-score, though it achieved the highest user accuracy in Rwanda (0.87 ± 0.06) and Zambia (0.81 ± 0.06).

Regional and map-specific performance patterns

Countries showed substantial performance variation, revealing important regional differences in classification challenges and the varying suitability of different mapping approaches across agricultural contexts.

Rwanda achieved the strongest cropland classification results, demonstrating the highest mean user accuracy (0.72 ± 0.07) and producer accuracy (0.60 ± 0.15) across all maps. The relatively strong performance across multiple products in Rwanda may reflect the country's intensive agricultural land use (56% cropland fraction, f in Table 4) and relatively homogeneous farming systems, which facilitate satellite-based crop detection. Digital Earth Africa (DEA) and the ensemble map performed particularly well in this context, with the ensemble achieving the highest overall accuracy (0.67 ± 0.03).

Malawi maps showed notably lower user accuracy (0.51 ± 0.07) and producer accuracy (0.36 ± 0.20), despite a cropland fraction (21%). The poor performance may be attributed to Malawi's complex agricultural landscape, characterized by small, fragmented fields that challenge classification (Djenontin et al., 2022; Mungai et al., 2020). GLAD and DEA showed the strongest results here, with DEA achieving the highest producer accuracy (0.69 ± 0.04), suggesting these products better handle landscape heterogeneity.

Kenya's results highlight a common challenge in land cover mapping: while the country achieved the highest overall accuracy (0.90), it showed relatively low user accuracy for cropland (0.47). This pattern reflects Kenya's sparse cropland distribution (9% cropland fraction), where high overall accuracy can be driven by correctly classifying the dominant non-crop class, while cropland classification remains challenging. The low cropland fraction makes commission errors particularly impactful on user accuracy metrics. Esri LULC performed best here for user accuracy (0.62 ± 0.08), while Copernicus achieved the highest producer accuracy (0.70 ± 0.06), though with lower user accuracy (0.42 ± 0.07), illustrating the precision–recall trade-off.

Tanzania and Uganda showed intermediate performance levels, with Tanzania achieving higher overall accuracies (mean OA = 0.81 ± 0.03) than Uganda (mean OA = 0.69 ± 0.04). In both countries, GLAD demonstrated balanced performance across metrics, achieving the highest F1 Scores and competitive user and producer accuracies. This suggests GLAD's classification approach is well-suited to the mixed agricultural-pastoral landscapes common in these regions.

Zambia exhibited interesting patterns, with relatively high user accuracies (mean UA = 0.72 ± 0.07) but moderate producer accuracies (mean PA = 0.46 ± 0.23). DEA and GLAD both achieved strong, balanced performance here (both with F1-scores > 0.71), while the Dynamic-World map achieved the highest user accuracy (0.81 ± 0.04) at the cost of lower producer accuracy (0.30 ± 0.06).

These accuracy results, particularly the PA and UA values, directly underpin the analysis in Section 4.2, where we evaluate the relative efficiency of each map for stratified cropland area estimation. Following Skakun (2025), the interplay between PA, UA, and each country's cropland fraction determines which map minimizes the sampling effort required to achieve a target precision, making the map-specific accuracy patterns documented above a critical input rather than a standalone evaluation.

4.2. Relative efficiency of maps for area estimation

Fig. 3 presents the relative efficiency matrices $\eta^k_{A,B}$ for each country, comparing all land cover maps for stratification in cropland area estimation. We computed the metrics using Eq. (4) at a target coefficient of variation of $\gamma = 0.1$ (10%), based on each country's cropland fraction f (Table 4). The relative efficiency $\eta^k_{A,B}$ quantifies how many times more efficient Map A is compared to Map B for estimating the area of class k; values greater than 1 indicate that Map A requires fewer samples to achieve equivalent precision (Skakun, 2025).

4.2.1. Producer's and user's accuracy contributions to map efficiency

Following the framework established by Skakun (2025), we analyze how classification accuracy metrics, specifically producer's accuracy (PA) and user's accuracy (UA), impact sampling efficiency for area estimation. Skakun (2025) demonstrated that for minor classes (true proportion $f < 0.5$), PA has a greater impact on relative efficiency than UA, whereas for major classes ($f > 0.5$), UA becomes the dominant factor. Our empirical results across six African countries strongly confirm these analytical findings.

Minor class efficiency patterns (Kenya, Malawi, Tanzania, Uganda, Zambia). Five of our six study countries exhibit minor cropland classes (f = 0.09–0.32), providing a robust test of Skakun (2025)'s theoretical framework for rare-class scenarios. In all five cases, the most efficient maps achieved substantially higher PA than the least efficient maps, with PA differences consistently exceeding UA differences. This pattern is particularly important for applications such as land-cover change monitoring, where omission errors for rare land change classes in large, stable strata can introduce substantial uncertainty into area estimates (Olofsson et al., 2020).

Table 3
Cropland classification accuracy metrics by country, evaluated against 2019 reference data. Best performance for each metric-country pair (in each row) is highlighted in **bold blue**, with the second-best performance in **bold black**.

Country	Metric	Ensemble	DEA	GLAD	Esri LULC	DynamicWorld	GLCFCS30	Copernicus	Mean
Kenya	Precision (UA)	**0.619 ± 0.08**	0.398 ± 0.08	0.575 ± 0.08	**0.624 ± 0.08**	0.249 ± 0.08	0.373 ± 0.07	0.419 ± 0.07	0.465 ± 0.13
	Recall (PA)	0.502 ± 0.08	**0.651 ± 0.06**	0.630 ± 0.07	0.583 ± 0.08	0.467 ± 0.05	0.376 ± 0.07	**0.695 ± 0.06**	0.558 ± 0.11
	F1 Score	0.555 ± 0.24	0.494 ± 0.20	**0.602 ± 0.23**	**0.603 ± 0.24**	0.325 ± 0.18	0.374 ± 0.22	0.523 ± 0.19	0.497 ± 0.10
	OA	**0.931 ± 0.01**	0.886 ± 0.02	0.928 ± 0.01	**0.934 ± 0.01**	0.833 ± 0.02	0.892 ± 0.01	0.891 ± 0.02	0.899 ± 0.03
Rwanda	Precision (UA)	**0.764 ± 0.04**	0.673 ± 0.04	0.698 ± 0.05	0.761 ± 0.05	**0.867 ± 0.04**	0.623 ± 0.04	0.681 ± 0.04	0.724 ± 0.07
	Recall (PA)	0.596 ± 0.05	**0.797 ± 0.04**	0.580 ± 0.04	0.527 ± 0.05	0.286 ± 0.06	0.678 ± 0.04	**0.722 ± 0.04**	0.598 ± 0.15
	F1 Score	0.670 ± 0.13	**0.730 ± 0.11**	0.634 ± 0.13	0.622 ± 0.14	0.430 ± 0.13	0.649 ± 0.13	**0.701 ± 0.12**	0.634 ± 0.09
	OA	**0.670 ± 0.03**	**0.669 ± 0.03**	0.623 ± 0.03	0.641 ± 0.03	0.574 ± 0.03	0.589 ± 0.03	0.653 ± 0.03	0.631 ± 0.04
Malawi	Precision (UA)	0.567 ± 0.04	0.471 ± 0.04	**0.573 ± 0.04**	**0.583 ± 0.04**	0.560 ± 0.03	0.376 ± 0.03	0.447 ± 0.04	0.511 ± 0.07
	Recall (PA)	0.224 ± 0.07	**0.692 ± 0.04**	**0.516 ± 0.04**	0.308 ± 0.06	0.120 ± 0.09	0.158 ± 0.06	0.514 ± 0.04	0.362 ± 0.20
	F1 Score	0.322 ± 0.13	**0.561 ± 0.11**	**0.543 ± 0.13**	0.403 ± 0.14	0.197 ± 0.12	0.222 ± 0.12	0.478 ± 0.13	0.389 ± 0.14
	OA	0.802 ± 0.01	0.773 ± 0.02	**0.819 ± 0.01**	**0.809 ± 0.01**	0.796 ± 0.01	0.769 ± 0.02	0.765 ± 0.02	0.791 ± 0.02
Tanzania	Precision (UA)	**0.701 ± 0.04**	0.557 ± 0.04	0.666 ± 0.04	**0.728 ± 0.04**	0.658 ± 0.04	0.390 ± 0.04	0.508 ± 0.04	0.601 ± 0.11
	Recall (PA)	0.390 ± 0.05	**0.658 ± 0.04**	0.651 ± 0.04	0.287 ± 0.06	0.295 ± 0.06	0.322 ± 0.04	**0.676 ± 0.04**	0.469 ± 0.17
	F1 Score	0.501 ± 0.13	**0.603 ± 0.12**	**0.659 ± 0.12**	0.412 ± 0.13	0.408 ± 0.13	0.353 ± 0.12	0.580 ± 0.11	0.502 ± 0.11
	OA	**0.835 ± 0.01**	0.816 ± 0.02	**0.856 ± 0.01**	0.825 ± 0.01	0.817 ± 0.02	0.748 ± 0.02	0.792 ± 0.02	0.813 ± 0.03
Uganda	Precision (UA)	**0.586 ± 0.03**	0.499 ± 0.03	**0.587 ± 0.03**	0.543 ± 0.03	0.579 ± 0.03	0.418 ± 0.03	0.508 ± 0.03	0.531 ± 0.06
	Recall (PA)	0.519 ± 0.04	**0.787 ± 0.03**	**0.824 ± 0.03**	0.372 ± 0.04	0.189 ± 0.06	0.578 ± 0.03	0.662 ± 0.03	0.562 ± 0.21
	F1 Score	0.551 ± 0.11	**0.611 ± 0.08**	**0.686 ± 0.08**	0.441 ± 0.11	0.285 ± 0.10	0.485 ± 0.09	0.575 ± 0.09	0.519 ± 0.12
	OA	**0.726 ± 0.02**	0.675 ± 0.02	**0.755 ± 0.02**	0.695 ± 0.02	0.693 ± 0.02	0.603 ± 0.02	0.683 ± 0.02	0.690 ± 0.04
Zambia	Precision (UA)	**0.789 ± 0.04**	0.683 ± 0.04	0.664 ± 0.04	0.785 ± 0.04	**0.812 ± 0.04**	0.661 ± 0.03	0.612 ± 0.04	0.715 ± 0.07
	Recall (PA)	0.301 ± 0.05	**0.794 ± 0.04**	**0.783 ± 0.04**	0.332 ± 0.05	0.274 ± 0.06	0.168 ± 0.08	0.531 ± 0.05	0.455 ± 0.23
	F1 Score	0.436 ± 0.12	**0.734 ± 0.11**	**0.719 ± 0.11**	0.466 ± 0.13	0.410 ± 0.12	0.269 ± 0.12	0.569 ± 0.14	0.515 ± 0.16
	OA	0.799 ± 0.02	**0.852 ± 0.02**	**0.842 ± 0.02**	0.804 ± 0.02	0.797 ± 0.02	0.763 ± 0.02	0.792 ± 0.02	0.807 ± 0.03
Mean	Precision (UA)	**0.671 ± 0.04**	0.547 ± 0.04	0.627 ± 0.05	**0.671 ± 0.05**	0.621 ± 0.04	0.473 ± 0.04	0.529 ± 0.05	
	Recall (PA)	0.422 ± 0.06	**0.730 ± 0.04**	**0.664 ± 0.05**	0.402 ± 0.06	0.272 ± 0.06	0.380 ± 0.06	0.633 ± 0.04	
	F1 Score	0.506 ± 0.14	**0.622 ± 0.12**	**0.640 ± 0.14**	0.491 ± 0.15	0.342 ± 0.13	0.392 ± 0.13	0.571 ± 0.13	
	OA	0.794 ± 0.02	0.778 ± 0.02	**0.804 ± 0.02**	0.785 ± 0.02	0.752 ± 0.02	0.727 ± 0.02	0.763 ± 0.02	

Legend: DEA = Digital Earth Africa, OA = Overall Accuracy, PA = Producer's Accuracy, UA = User's Accuracy.

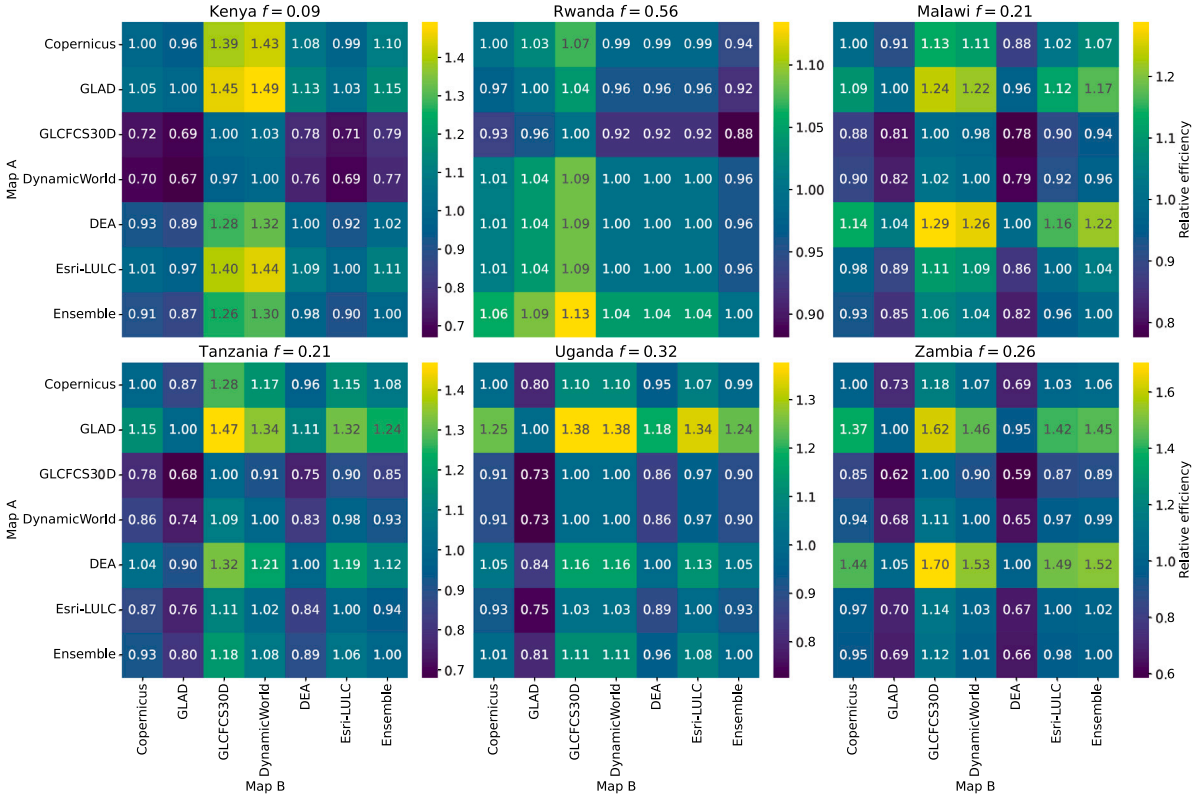


Fig. 3. Pairwise relative efficiency matrices ($\eta^{k}_{A,B}$) of land cover maps for cropland area estimation in each country. Each cell compares the efficiency of Map A (row) to Map B (column) under a stratified random sampling design. We calculated efficiency based on the optimal sample size required to reach a target coefficient of variation $\gamma = 0.1$, using each country's cropland area proportion (f). Values greater than 1 indicate that Map A requires fewer samples to achieve the same estimation precision as Map B.

Table 4
Summary of optimal sampling strategies for cropland area estimation across countries. For each country, we identify the most efficient land cover map and corresponding sample design to achieve a 10% coefficient of variation. The table shows the proportion of cropland area (f), the total sample size required (n_{opt}), the Neyman-optimal allocation between strata, and efficiency gains relative to median and least efficient maps.

Country	Map	f	Sample size	Strata allocation		Efficiency gain (%)	
			n_{opt}	Crop	Non-crop	vs Median	vs Least
Kenya	GLAD	0.09	616	134	482	11.5	33.0
Rwanda	Ensemble	0.56	67	27	40	4.3	11.8
Malawi	DEA	0.21	288	125	163	14.0	22.4
Tanzania	GLAD	0.21	239	71	168	19.5	31.9
Uganda	GLAD	0.32	146	84	62	19.8	27.4
Zambia	DEA	0.26	159	68	91	32.9	41.3

Efficiency Gain measures percentage reduction in sample size: **vs Median** compares against the median-efficiency map (conservative estimate of typical gains), **vs Least** compares against the least efficient map (maximum potential gain). Median-efficiency maps: Digital Earth Africa (Kenya), Esri LULC (Rwanda, Malawi, Zambia), Ensemble (Tanzania), Copernicus (Uganda). The least efficient map is GFLFC30 for all countries except Kenya (DynamicWorld).

- **Uganda** ($f = 0.32$): GLAD (most efficient) achieved PA = 0.82, UA = 0.59, while DynamicWorld (least efficient) had PA = 0.19, UA = 0.58. The critical difference is ΔPA = +0.63 versus ΔUA = +0.01, demonstrating that the 63% point gain in PA, despite nearly identical UA, reduced sample requirements from 201 to 146 (27% reduction).
- **Zambia** ($f = 0.26$): DEA (most efficient, PA = 0.79, UA = 0.68) outperformed GFLFC30 (least efficient, PA = 0.17, UA = 0.66) primarily through a 62% point PA advantage (ΔPA = +0.63) with minimal UA difference (ΔUA = +0.02), reducing sample size from 271 to 159 (41% reduction).
- **Malawi** ($f = 0.21$): DEA achieved the highest efficiency (PA = 0.69, UA = 0.47) compared to GFLFC30's (PA = 0.16, UA = 0.38). Again, PA improvement (ΔPA = +0.54) far exceeded UA improvement (ΔUA = +0.10), yielding a 22% sample size reduction (371 to 288).
- **Tanzania** ($f = 0.21$): GLAD (PA = 0.65, UA = 0.67) proved more efficient than GFLFC30 (PA = 0.32, UA = 0.39), with ΔPA = +0.33 exceeding ΔUA = +0.28, resulting in a 32% efficiency gain.

These patterns align precisely with the theoretical framework of Skakun (2025): when the target class is rare, omission errors (1 – PA) disproportionately increase the variance of the area estimator because they reduce the effective sample size in the already-sparse cropland stratum. Maps that minimize omission through high PA therefore yield superior stratification efficiency, even when UA is moderate or when the PA increase comes at the expense of UA (Skakun, 2025).

Unexpected pattern: Kenya. Kenya ($f = 0.09$, the sparsest cropland) presents an apparent exception. GLAD (most efficient, PA = 0.63, UA = 0.58) outperformed DynamicWorld (least efficient, PA = 0.47, UA = 0.25), but here ΔUA = +0.33 exceeded ΔPA = +0.16. This deviation is explained by DynamicWorld's exceptionally low UA (0.25), which creates extremely high commission error (75%). At such extreme values, even for minor classes, commission errors inflate the non-cropland stratum variance sufficiently to degrade efficiency. Skakun (2025)'s framework accommodates this: while PA generally dominates for minor classes, the relationship is nonlinear, and extreme imbalances in either direction reduce efficiency (Skakun (2025), Fig. 9-10). The key insight remains valid: GLAD's more balanced accuracy (PA/UA = 1.10) proved more efficient than DynamicWorld's severe imbalance (PA/UA = 1.88).

Major class efficiency pattern: Rwanda. Rwanda ($f = 0.56$) represents the sole major class in our dataset, providing a test of Skakun (2025)'s study that UA should dominate efficiency for abundant classes. The ensemble map (most efficient, PA = 0.60, UA = 0.76) outperformed GFLFC30 (least efficient, PA = 0.68, UA = 0.62). Here, the UA advantage (ΔUA = +0.14) indeed exceeded the PA difference (ΔPA = −0.08), confirming Skakun (2025)'s framework: when the target class is abundant, commission errors (1 – UA) primarily determine stratum variance because the cropland stratum dominates sample allocation. Interestingly, the ensemble map achieved higher efficiency despite *lower* PA, demonstrating that for major classes, prioritizing UA over PA yields better stratification.

4.2.2. Efficiency gains and practical implications

Table 4 summarizes the optimal sampling strategy for each country, including sample size requirements and allocation between cropland and non-cropland strata using Neyman allocation (Skakun, 2025; Cochran, 2002-01-03). We quantify efficiency gains using two complementary benchmarks: the median-efficiency map (a conservative estimate) and the least efficient map (the maximum potential gain). These dual metrics provide both practical guidance for typical map choices and bounds on the importance of optimal selection.

Efficiency gains relative to median map. Efficiency gains relative to the median map, representing typical product performance, ranged from 4.3% in Rwanda to 32.9% in Zambia (Table 4). These conservative estimates demonstrate that even when excluding the worst-performing maps, strategic map selection yields meaningful reductions in sampling effort.

Rwanda's modest gain (3 samples saved, from 70 to 67) reflects the compressed efficiency variation when cropland is abundant: all maps achieved relatively high UA (0.62–0.87), and the narrow sample size range (67–76) indicates that map choice matters less for major classes with balanced accuracies. In contrast, Zambia's substantial median-relative gain (78 samples saved, 32.9%) underscores the critical importance of map selection in moderate cropland-fraction contexts ($f = 0.26$). DEA's high PA (0.79) relative to the median map's (Esri LULC, PA = 0.33) drove this large efficiency differential, consistent with Skakun (2025)'s framework for minor classes.

Intermediate cases showed median-relative gains of 11–20%: Kenya (11.5%, 80 samples), Malawi (14.0%, 47 samples), Tanzania (19.5%, 58 samples), and Uganda (19.8%, 36 samples). These gains directly translate to reduced field data collection or image interpretation costs—at 5–15 min per sample unit, savings range from 4–20 h of analyst time per country.

Maximum efficiency gains relative to least efficient map. Maximum efficiency gains relative to the least efficient map ranged from 11.8% (Rwanda) to 41.3% (Zambia), illustrating the full range of potential sample size reductions through optimal stratification (Table 4). GFLFC30 emerged as the least efficient map across five of six countries, consistently exhibiting low PA (0.16–0.38), which undermines stratification effectiveness for minor classes.

In absolute terms, Kenya demonstrated the largest potential savings (303 samples), reducing requirements from 919 (DynamicWorld) to 616 (GLAD)—a difference attributable to GLAD's $\Delta PA = +0.16$ and $\Delta UA = +0.33$ advantage. Tanzania achieved the second-largest absolute savings (112 samples), while Rwanda's high cropland fraction limited maximum savings to 9 samples despite a 12% relative gain.

4.2.3. Implications for map producers

Our results provide actionable guidance for both map producers developing new cropland products and map users selecting among existing products for area estimation:

For map producers. When the intended application is area estimation of rare or minor classes (the typical scenario for cropland in Africa), algorithm optimization should prioritize producer's accuracy over user's accuracy. Gains in PA yield larger improvements in sampling efficiency than equivalent gains in UA for minor classes. For major classes, this priority reverses: when cropland dominates the landscape, UA becomes

Table 5
National cropland area estimates with uncertainties (±1 standard error) based on samples from validation campaign. Results use the most efficient land-cover map for each country and report the achieved coefficient of variation (CV). The pixel count estimators are included for comparison.

Country	Map	Area (Mha)	SE (Mha)	CV (%)	Pixel count (Mha)
Kenya	GLAD	4.40	0.43	9.65	5.83
Rwanda	Ensemble	1.40	0.15	10.76	1.16
Malawi	DEA	3.63	0.29	8.03	3.78
Tanzania	GLAD	12.66	1.61	12.71	15.75
Uganda	GLAD	6.14	0.76	12.35	5.83
Zambia	DEA	6.31	0.93	14.67	8.99

the critical metric. Importantly, severe imbalances in either direction (PA/UA ratios > 1.5 or < 0.67) reduce efficiency, suggesting that balanced accuracy, even if both metrics are moderate, may outperform high accuracy in one metric paired with poor accuracy in the other.

4.3. Sample-based cropland area estimates and uncertainty

Using the labeled reference points generated through stratified random sampling (Table 4) and the stratified estimator (Eq. (6)), we derived unbiased national-level estimates of cropland area for each study country in 2019. Table 5 presents the sample-based cropland area estimates, associated standard errors, and resulting coefficients of variation (CV), alongside pixel-counting estimates from the maps for comparison. Fig. 4 displays the error matrices in terms of estimated area proportions, following recommended good practices (Stehman and Foody, 2019; Olofsson et al., 2014).

We achieved CV values ranging from 8.03% (Malawi) to 14.67% (Zambia), with two countries (Kenya, Malawi) meeting the ambitious 10% precision target. The four countries that slightly exceeded this target, Rwanda (10.76%), Tanzania (12.71%), Uganda (12.35%), and Zambia (14.67%), nonetheless demonstrate operationally precise area estimates suitable for agricultural monitoring and decision-making.

To contextualize these results, CVs of 8%–15% align with theoretical expectations for stratified random sampling. Established studies (Skakun, 2025; Olofsson et al., 2014; Cochran, 2002-01-03) demonstrate that stratified sampling achieves 1.5–3.3× greater efficiency than simple random sampling for rare-class estimation, even with moderately accurate maps (PA and UA ≥ 0.6). Additionally, our precision substantially exceeds typical values from multi-product ensemble approaches: Tubiello et al. (2023a) report relative uncertainties of 29%–38% (equivalent to CVs of approximately 27%–36%) for map-based cropland estimates in Eastern and Southern Africa, reflecting high disagreement among six independent land cover products. While two countries met the 10% target, four exceeded it modestly (10.76–14.67%). These slight exceedances reflect residual classification errors evident in the error matrices (Fig. 4). Despite using the most efficient maps, residual omission and commission errors contribute to within-stratum variance, for example, Zambia's DEA map exhibits PA = 0.88 and UA = 0.62, meaning 12% cropland omission and 38% commission error inflate stratum variance beyond the theoretical minimum achievable with perfect classification (Skakun, 2025).

4.4. Comparison of sample-based with agricultural stats

We compared our 2019 sample-based cropland area estimates against official land use statistics from three sources: FAOSTAT, the USDA Foreign Agricultural Service (FAS), and national government agencies (Section 2.3, Table 1).

The sample-based estimates showed the strongest correlation with FAOSTAT arable land data (Table 6; R^2=0.89, NRMSE = 10.3%), despite our reference data definition theoretically aligning with the broader FAOSTAT cropland category. The correlation with FAOSTAT

Table 6
Results of the comparison between the country statistics from the sample-based cropland estimates and the agricultural land use statistics from different sources (R^2 and NRMSE). Scatter plots in Appendix C.

Statistics sources	R^2	NRMSE (%)
FAOSTAT Cropland	0.85	15.1
FAOSTAT Arable land	0.89	10.3
Country official report	0.82	17.2
USDA-FAS	0.83	55.0

Note: All comparisons include 6 countries except the official report (5 countries) due to unavailable Malawi data.

cropland statistics was slightly lower but still strong (R^2=0.85, NRMSE = 15.1%). This pattern mirrors findings from Tubiello et al. (2023b), who demonstrated that remote sensing products, even those nominally including permanent crops, often achieve closer statistical alignment with arable land than with cropland, potentially due to better detection of annual herbaceous crops or to the exclusion of temporary meadows and pastures that FAOSTAT includes in arable land but remote sensing typically classifies as grassland.

Comparable R^2 values were observed with national official reports (excluding Malawi due to data unavailability) (R^2 = 0.82, NRMSE = 17.2%). In contrast, USDA-FAS figures, while showing similar R^2 (0.83), exhibited substantially higher NRMSE (55.0%), reflecting that these data represent the harvested area of major commodity crops rather than the total cropland extent (Fig. C.1).

Fig. 5 illustrates the relationship between our estimates and various statistical sources for each country. For Malawi, Tanzania, and Uganda, the FAOSTAT arable land figures fall within one standard error ($\pm 1SE$) of our estimates, indicating good agreement. Rwanda shows similar consistency with FAOSTAT cropland values. For Kenya, our estimate aligns well with both FAOSTAT metrics and official statistics, though the USDA-FAS figure is notably lower, likely reflecting that agency's focus on major grain crops.

Zambia presents an interesting case where our estimate (6.3 ± 0.9 Mha) substantially exceeds all official statistical sources (ranging from 1.4 to 3.8 Mha). This discrepancy may reflect actual cropland area missed by national surveys in heterogeneous landscapes, or potential commission errors in our stratification map leading to overestimation, or differences in how fallow land and marginal agriculture are classified across data sources (Liu et al., 2018; Gallego et al., 2014).

Rather than treating remote sensing and traditional statistics as competing approaches, future efforts should actively integrate them as complementary sources. Remote sensing provides temporal consistency and spatial comprehensiveness that periodic surveys cannot match, while national surveys supply the contextual detail and institutional knowledge that purely satellite-derived estimates lack. To realize this integration in practice, three steps are essential: first, harmonizing crop and land-use definitions across remote sensing products and national statistical frameworks; second, establishing validation protocols that benchmark satellite-derived estimates against survey data at subnational scales; and third, embedding these hybrid workflows within national statistical agencies so that the resulting data carry official credibility. Without such structured collaboration, the two data streams will continue to develop in parallel—each valuable on its own, but falling short of the reliability that agricultural policy decisions demand.

5. Discussion

Our comparative evaluation of seven cropland maps demonstrates that strategic selection based on producer's and user's accuracy balance reduces sampling costs by 20%–40% while achieving 8%–15% precision in area estimates. These empirical results validate Skakun (2025)'s theoretical framework and provide actionable guidance for map users and producers in operational agricultural monitoring systems.



Fig. 4. Area proportion error matrices showing classification performance for the most efficient cropland map in each country. Matrices display the relationship between reference data (columns) and map predictions (rows) as proportional areas.

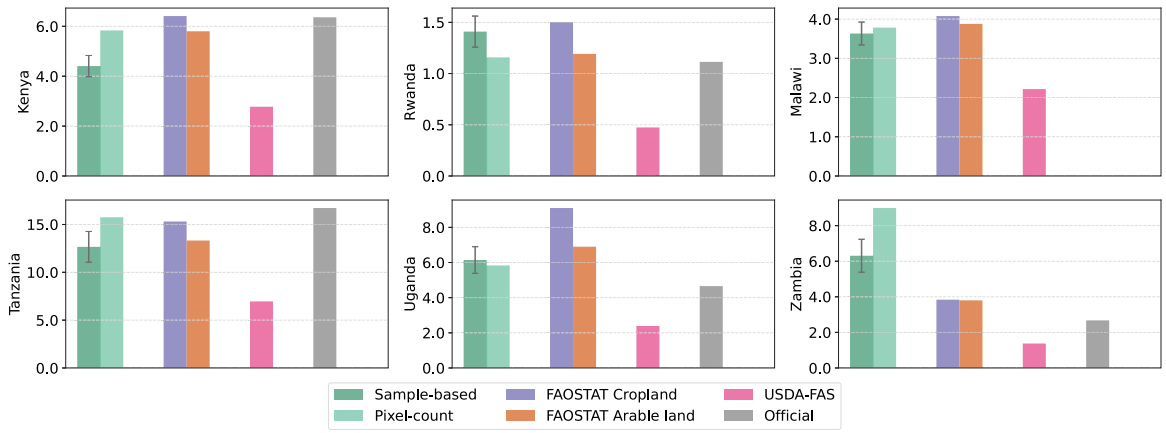


Fig. 5. Cross-country comparison of cropland area estimates (Mha) from multiple sources. Error bars represent the standard errors of sample-based estimates. Sources include remote sensing maps, FAOSTAT data, USDA-FAS statistics, and national government reports.

5.1. Strategic guidance for map selection and algorithm optimization

Practitioners selecting among existing maps for stratified area estimation should prioritize stratification efficiency over traditional accuracy metrics, with selection depending critically on target class proportion. For rare cropland ($f < 0.5$), high producer's accuracy (PA) minimizes omission from the sparse cropland stratum. In five countries with minor cropland, the most efficient maps exhibited substantially higher PA than alternatives; for example, Uganda's GLAD map (PA = 0.70) required 27% fewer samples than the least efficient option, despite a moderate UA = 0.74. The practical decision rule: for $f < 0.25$, select maps with PA $\geq$ 0.70 even if UA is moderate (0.55–0.65). Our empirical results confirm Skakun (2025) analysis that PA dominates efficiency for rare classes, with Zambia (PA = 0.88, UA = 0.62) achieving 41% efficiency gain despite unbalanced accuracy.

Conversely, when cropland is abundant ($f \geq 0.5$), user's accuracy (UA) should be prioritized. Rwanda ($f = 0.55$) demonstrated this pattern: the Ensemble map (UA = 0.70, PA = 0.58) proved most efficient despite lower PA than alternatives. For $f \geq 0.5$, select maps with UA ≥ 0.70, accepting moderate PA (0.55–0.65) as the efficiency-maximizing trade-off. Regardless of cropland fraction, maps with PA/UA ratios > 1.5 or < 0.67 consistently showed reduced efficiency. Kenya's DynamicWorld (PA/UA = 1.88) and Rwanda's DynamicWorld (PA/UA = 0.33) both ranked among the least efficient despite reasonable overall accuracy. Critically, overall accuracy (OA) and F1-score fail as efficiency predictors; maps with identical F1-scores exhibited efficiency differences exceeding 50% in our analysis.

The essential workflow for map users involves five steps: (1) estimate target cropland fraction from available data or pilot sampling, (2) obtain PA and UA for candidate maps, (3) prioritize high PA if $f < 0.5$ or high UA if $f \geq 0.5$, (4) eliminate maps with PA/UA ratios > 1.5 or < 0.67, and (5) conduct pilot sampling (50–100 points) to estimate relative efficiency empirically if multiple viable maps remain. At 3–5 min per sample for satellite-based interpretation using multitemporal imagery, this pilot assessment requires 2.5–8 h of analyst time, a modest investment that can save thousands of samples (and associated costs) in operational surveys.

Map producers whose products target area estimation should fundamentally rethink algorithm optimization priorities. For rare-class applications typical of cropland mapping, algorithm development should maximize PA while accepting moderate UA. Traditional machine learning optimization (maximizing OA or F1-score) systematically underweights rare-class PA because the abundance of non-cropland pixels dominates model training (Skakun, 2025; Waldner et al., 2017; He and Garcia, 2009). For cropland mapping in regions with $f < 0.25$ (PA priority), aim for PA ≥ 0.75 even if UA drops to 0.60–0.65. Conversely, for regions with $f > 0.5$ (UA priority), reverse this priority to maximize UA. Our results show that GLAD and DEA's binary classification focus yields superior efficiency (33%–41% gains) compared to multi-class products. For abundant cropland regions ($f > 0.5$), reverse this priority by maximizing UA while accepting moderate PA.

Producers should adopt relative efficiency (Skakun, 2025, Eq. 15) as the primary benchmark when area estimation is the intended application. Kenya's GLAD vs DynamicWorld comparison illustrates this: while traditional metrics show 9.5% OA difference, GLAD requires 33% fewer samples (616 vs 919), saving 303 sample interpretations per survey cycle. At typical satellite interpretation rates of 3–5 min per sample, this represents 15–25 h of saved analyst time per cycle. For comparison, if reference data were collected via field surveys at costs similar to European LUCAS operations (€35–40/sample, inflation-adjusted from Gallego (2012)), the savings would be €7600–10,600 per cycle, though direct field survey cost data for sub-Saharan Africa remains limited in the literature (d'Andrimont et al., 2020). Reporting efficiency gains alongside accuracy metrics communicates real-world value and justifies algorithm development investments.

Our analysis reveals substantial geographic variability in map performance: no single product maintained consistently superior accuracy across all six countries, challenging the 'one-size-fits-all' paradigm in global cropland mapping. This variability stems from multiple factors, including the inherent complexity of smallholder-dominated agricultural landscapes, geographic bias in the selection of training data, with many products under-representing African contexts, and oversimplified classification schemes. These findings underscore two critical implications: (1) practitioners must conduct region-specific validation before operational deployment, and (2) global products should be evaluated not by their peak performance but by their consistency across diverse agricultural systems (H. Kerner et al., 2024).

Our cross-product comparison revealed that binary-focused products (GLAD, DEA) consistently outperformed multi-class maps (DynamicWorld, ESRI) in cropland accuracy assessment and stratification efficiency, suggesting that the composition of the training dataset is critical. Producers should ensure that cropland prevalence in the training data matches or exceeds the true prevalence and prioritize geographically diverse cropland samples to minimize omission errors. The same principle applies to crop-type mapping: training data for rare or regionally important crops should be deliberately oversampled to improve their producer's accuracy and, consequently, overall stratification efficiency for crop-specific area estimation.

5.2. Policy implications for food security monitoring

Our findings have direct implications for food security policy and planning at both national and regional levels. The improvements in sampling efficiency we demonstrate, reducing required sample sizes by 20%–40% across the evaluated countries, enable resource-constrained governments to implement more comprehensive and frequent agricultural monitoring within existing budgets. This enhanced monitoring capacity directly supports evidence-based policy decisions for food security early warning systems, agricultural investment planning, and climate adaptation strategies. The framework we present can be integrated directly into traditional agricultural statistics collection processes. National agricultural surveys and census operations can adopt stratified sampling using remote-sensing-derived cropland maps to improve efficiency without disrupting established data collection protocols. This integration allows countries to modernize their monitoring systems by incorporating satellite-derived stratification while maintaining continuity with existing statistical frameworks and historical data series—addressing a critical policy challenge for statistical offices, balancing innovation with consistency. For policymakers managing food security in rapidly changing agricultural landscapes, the improved precision of area estimates (achieving coefficients of variation between 8%–15%) provides the statistical confidence necessary for strategic planning and resource allocation decisions. Furthermore, the standardized approach demonstrated across diverse African contexts enables regional coordination of food security policies and supports continental initiatives, such as the Comprehensive Africa Agriculture Development Programme (CAADP), which require consistent and comparable agricultural statistics across member countries.

5.3. Limitations

This study has several methodological constraints that warrant consideration. Labeling reference data from satellite imagery introduces interpreter variability, particularly in heterogeneous smallholder landscapes with mixed cropping systems and indistinct field boundaries (Stehman et al., 2022). While we used high-resolution PlanetScope imagery (3 m) through NASA's Commercial Smallsat Data Acquisition Program (Neigh et al., 2013) and established consensus protocols to resolve classification disputes, we did not formally quantify interpreter variability through repeated independent labeling (Stehman et al., 2022). Although consensus was reached on all reference labels, this does not eliminate uncertainty in borderline cases where classification is genuinely ambiguous or potential systematic interpreter bias. The proprietary nature of PlanetScope data limits reproducibility; future studies should explore openly accessible alternatives such as Sentinel-2 (10 m) and Google Earth Pro imagery, though reduced spatial resolution may affect boundary delineation accuracy in fragmented smallholder systems.

Our analysis covers six countries representing diverse agricultural systems, from intensive smallholder cultivation in Rwanda and Malawi to extensive mixed crop-livestock systems in Tanzania and Zambia, but does not capture the full range of African agricultural contexts. Notably absent are North African cereal systems, West African tree crop zones, and Southern African large-scale commercial agriculture. While the Skakun (2025) framework we validate is theoretically general, empirical confirmation of PA/UA efficiency relationships in these

other contexts would strengthen confidence in extrapolation. Our findings regarding specific product performance should be interpreted as snapshots of 2019 product versions rather than permanent rankings, though the framework for efficiency evaluation (Section 4.2) remains applicable to newer products.

Our comparison of products spanning 10 m to 100 m resolution introduces scale-dependent uncertainties. Although our reference data (3 m PlanetScope imagery) are finer than all evaluated products, consistent with good practice recommendations (Olofsson et al., 2014), the accuracy metrics derived from point-level (10 × 10 m) reference labels may not equivalently reflect map quality across resolutions. A 100 m Copernicus pixel represents an area containing 100 of our reference units, so accuracy evaluation at the point level tests whether the coarse pixel's classification matches a small fraction of the area it represents, which may not fully capture that product's performance at its intended scale (Stehman and Wickham, 2011). Similarly, the wide range of country sizes in our study (from $26,000 km^2$ for Rwanda to $945,000 km^2$ for Tanzania) means that resolution-landscape interactions may differ across contexts: the same product resolution may capture cropland patterns differently in small, intensively cultivated landscapes versus large, heterogeneous ones. While H. Kerner et al. (2024) found only weak correlations between resolution and accuracy metrics across Sub-Saharan Africa, and our own results confirm that coarser products occasionally outperform finer ones (e.g., Copernicus in select cases, Table 3), these scale-related uncertainties should be considered when interpreting cross-product efficiency comparisons and when extrapolating our findings to other regions or spatial extents.

Finally, we report discrepancies between sample-based estimates and official national statistics, but cannot definitively determine which source better reflects reality. Official statistics often lack documentation of uncertainty and may reflect different definitions of cropland, temporal periods, or survey methodologies. Our sample-based estimates are design-unbiased and include quantified uncertainty, but this does not guarantee they are closer to true cropland extent than alternative sources. This fundamental validation challenge, common to all remote sensing area estimation studies, underscores the need for coordinated ground-based validation networks with standardized protocols, currently lacking across most of Africa.

These limitations suggest two priorities for future research: (1) extending the framework to multi-year assessments (e.g.,2024–2025) that characterize the temporal stability of map efficiency rankings and validate cropland change detection products, and (2) expanding geographic coverage to validate PA/UA efficiency relationships across the full diversity of African agricultural systems at provincial or regional scales. Future studies should adopt the unified sampling and estimation framework of Tyukavina et al. (2025), which provides design-based estimators for both static land cover and change products, to ensure consistent methodological implementation across different geographic extents, projection systems, and data sources.

6. Conclusion

This study presents an integrated framework that combines map accuracy assessment with relative efficiency analysis to optimize stratified sampling for cropland area estimation. Applied across six East and Southern African countries, our approach demonstrates that map selection should prioritize efficiency, defined as the ability to achieve target precision with minimal sample size, rather than classification accuracy alone. By quantifying how producer's and user's accuracies influence sampling efficiency, we provide operational guidance for selecting stratification maps that reduce reference data collection costs while maintaining statistical rigor.

No single land cover product performed optimally across all countries, confirming the necessity of country-specific evaluation. The most efficient maps, predominantly GLAD and Digital Earth Africa, enabled substantial reductions in required sample size (20%–40% fewer samples) compared to less efficient alternatives, translating directly into lower fieldwork and annotation costs. Our sample-based cropland area estimates achieved coefficients of variation between 8%–15%, a marked improvement over reported precision levels in the region (27%–36%), and showed strong agreement with FAOSTAT arable land statistics ($R^2 = 0.89$).

The framework's core contribution lies in operationalizing relative efficiency as a criterion for map selection. For rare classes such as cropland in sparsely cultivated landscapes, gains in producer's accuracy yield greater efficiency improvements than equivalent gains in user's accuracy. This insight guides algorithm developers and map producers: when area estimation is the primary objective, optimizing for the appropriate accuracy metric, aligned with the target class proportion, maximizes sampling efficiency and cost-effectiveness.

Our approach provides actionable guidance for national statistical agencies and agricultural monitoring programs, particularly in data-scarce regions where fieldwork resources are limited. The quantitative assessment of map efficiency informs decisions about which products warrant investment in reference data collection. Beyond immediate application, this framework supports temporal monitoring of cropland dynamics and enables reconciliation of satellite-based estimates with evolving agricultural statistics. By transparently communicating uncertainties and demonstrating how map quality affects estimation precision, we advance evidence-based agricultural decision-making across diverse farming systems worldwide.

CRediT authorship contribution statement

Adebowale Daniel Adebayo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Catherine Nakalembe:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Sergii Skakun:** Writing – review & editing, Supervision, Methodology, Investigation. **Diana Botchway Frimpong:** Data curation. **Alana Ginsburg:** Data curation. **Yash Hitesh Gadhiya:** Data curation. **Monica Adjei:** Data curation. **Hannah Kerner:** Writing – review & editing, Methodology. **Inbal Becker-Reshef:** Writing – review & editing, Funding acquisition.

Code and data availability

The data that support the findings of this study are available in **xylem-lab optimizing-cropland-area-est** GitHub Repository at https://github.com/xylem-lab/optimizing-cropland-area-est (Adebayo et al., 2025).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Operational definition of agricultural land use statistics

See Table A.1.

Table A.1
Operational definition of agricultural land use statistics.

Data sources	Definition
FAOSTAT	
Cropland	"Land used for cultivation of crops. The total of areas under Arable land and Permanent crops"
Arable land	"Land used for cultivation of crops in rotation with fallow, meadows and pastures within cycles of up to five years."
Permanent crops	"Land cultivated with long-term crops which do not have to be replanted for several years [...], land under trees and shrubs producing flowers [...]."
Temporary crops	"Land used for crops with a less-than-one-year growing cycle, which must be newly sown or planted for further production after the harvest."
Temporary meadows and pastures	"Land temporarily cultivated with herbaceous forage crops for mowing or pasture, as part of crop rotation periods of less than five years."
Temporary fallow	"Land that is not seeded for one or more growing seasons. The maximum idle period is usually less than five years."
USDA FAS	Land harvested of major commodity crops.

Table B.1
Description of the map data products evaluated. Detailed descriptions and definitions are from H. Kerner et al. (2024), with the exception of GLC_FCS30D, which is defined in Zhang et al. (2024).

Datasets	Resolution (m/px)	Coverage	Cropland definition
DEA	10	Continent	"A piece of land of minimum 0.01 ha that is sowed/planted and harvestable at least once within the 12 months after the sowing/planting date."
DynamicWorld	10	Global	"Human planted/plotted cereals, grasses, and crops."
Esri-LULC	10	Global	""Human planted/plotted cereals, grasses, and crops not at tree height; examples: corn, wheat, soy, fallow plots of structured land.""
Copernicus	100	Global	"Cultivated and managed vegetation/agriculture. Lands covered with temporary crops followed by harvest and a bare soil period (e.g., single and multiple cropping systems). Note that perennial woody crops will be classified as the appropriate forest or shrub land cover type."
GLAD	30	Global	"[...] land used for annual and perennial herbaceous crops for human consumption, forage (including hay), and biofuel. Perennial woody crops, permanent pastures, and shifting cultivation are excluded from the definition."
GLC_FCS30D	30	Global	"Rainfed cropland and Herbaceous cover cropland"

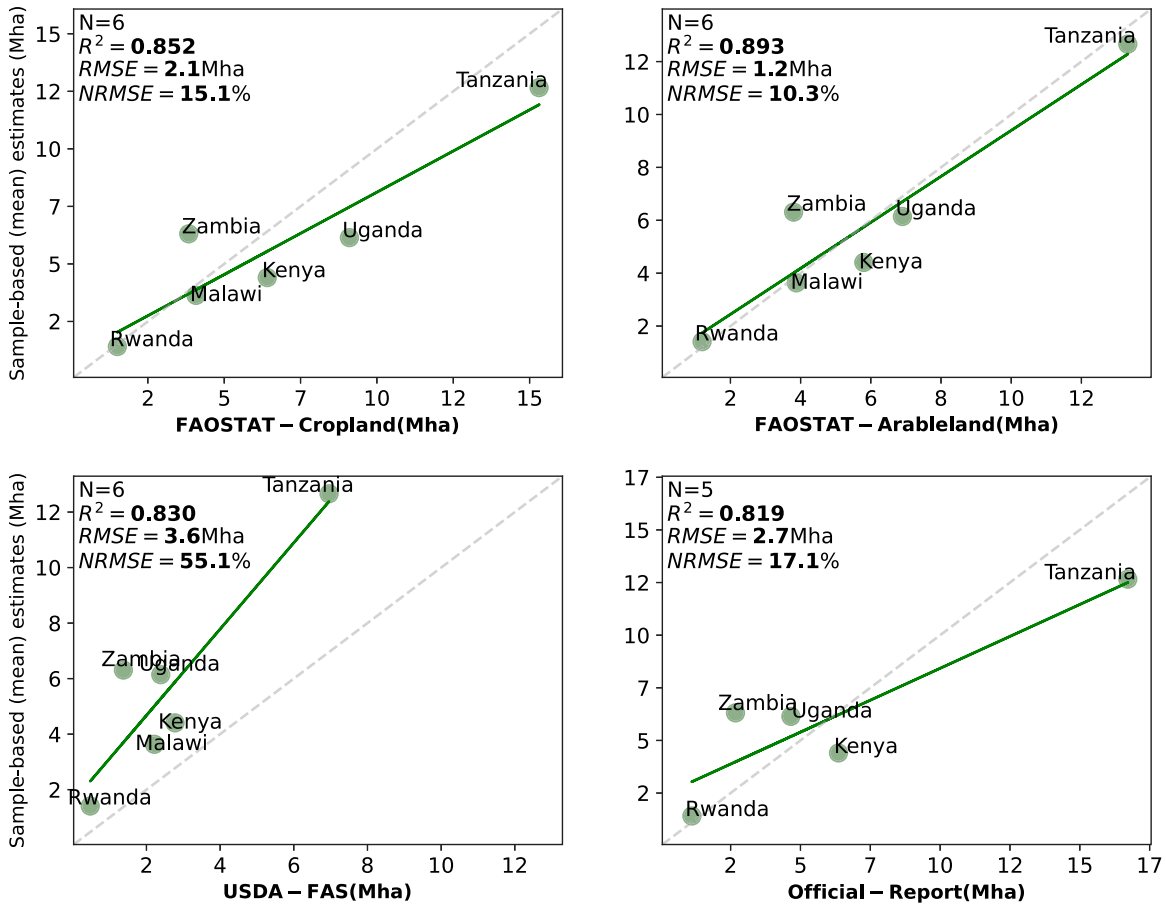


Fig. C.1. Scatter plots comparing sample-based cropland area estimates with agricultural statistics from four sources: (a) FAOSTAT Cropland category (includes permanent and temporary crops), (b) FAOSTAT Arable Land category (temporary crops only), (c) USDA Foreign Agricultural Service estimates, and (d) national official reports. Each point represents one country; the diagonal dashed line indicates perfect agreement (1:1). Statistical metrics quantify agreement: R^2 (coefficient of determination), RMSE (root mean squared error, Mha), and NRMSE (normalized RMSE as a percentage of the data range). Discrepancies arise from definitional inconsistencies (permanent vs. temporary crop inclusion), temporal misalignment (year-specific vs. multi-year aggregates), and methodological differences (design-based sampling vs. administrative reporting). Our estimates provide design-unbiased values with quantified uncertainty; official statistics lack documented error and are not independent ground truth.

Appendix B. Cropland definitions of map products

See Table B.1.

Appendix C. Comparison of sample-based estimates with agricultural statistics scatter plots

See Fig. C.1.

Data availability

All data and methods linked in paper.

References

Adebayo, A.D., Nakalembe, C., Skakun, S., Frimpong, D.B., Ginsburg, A., Gadhiya, Y.H., Adjei, M., Kerner, H., Becker-Reshef, I., 2025. Optimizing-cropland-area-est: Optimizing satellite-Based Cropland Area estimation through integrated map accuracy assessment and stratified sampling design across six African countries. https://github.com/xylem-lab/optimizing-cropland-area-est. GitHub repository; (Accessed 28 November 2025).

Agra, Comesa, 2022. Regional Food Balance Sheet. Alliance for Green Revolution in Africa and Common Market for Eastern and Southern Africa, URL: https://rfbsa.com. A digital platform that uses data from various public and private sources to develop near real-time and forward-looking food balance estimates across Eastern and Southern Africa.

APEC, 2013. Agricultural Statistics Best Practice Methodology Handbook. Technical Report, Asia-Pacific Economic Cooperation Technical Assistance and Training Facility Program, Singapore, URL: https://www.apec.org/docs/default-source/Publications/2013/12/Agricultural-Statistics-Best-Practice-Methodology-Handbook/2013_ATC_ag-stat-handbook.pdf.

Böttcher, H., Eisbrenner, K., Fritz, S., Kindermann, G., Kraxner, F., McCallum, I., Obersteiner, M., 2009. An assessment of monitoring requirements and costs of 'reduced emissions from deforestation and degradation'. Carbon Balance Manag. 4 (1), 7. http://dx.doi.org/10.1186/1750-0680-4-7.

Brown, C.F., Brumby, S.P., Guzder-Williams, B., Birch, T., Hyde, S.B., Mazzariello, J., Czerwinski, W., Pasquarella, V.J., Haertel, R., Ilyushchenko, S., Schwehr, K., Weisse, M., Stolle, F., Hanson, C., Guinan, O., Moore, R., Tait, A.M., 2022. Dynamic world, near real-time global 10 m land use land cover mapping | scientific data. Sci. Data 9 (1), 251. http://dx.doi.org/10.1038/s41597-022-01307-4, URL: https://www.nature.com/articles/s41597-022-01307-4. Number: 1.

Buchhorn, M., Lesiv, M., Tsendbazar, N.-E., Herold, M., Bertels, L., Smets, B., 2020. Copernicus global land cover layers—collection 2. Remote. Sens. 12 (6), 1044.

Burton, C., Yuan, F., Ee-Faye, C., Halabisky, M., Ongo, D., Mar, F., Addabor, V., Mamane, B., Adimou, S., 2022. Co-production of a 10-m cropland extent map for continental africa using sentinel-2, cloud computing, and the open-data-cube. Authorea Prepr. http://dx.doi.org/10.1002/essoar.10510081.1.

Cao, J., Zhang, Z., Luo, Y., Xu, J., Xie, J., Han, J., Tao, F., 2025. Mapping global yields of four major crops at 5-minute resolution from 1982 to 2015 using multi-source data and machine learning. Sci. Data 12, 357. http://dx.doi.org/10.1038/s41597-025-04650-4, URL: https://www.nature.com/articles/s41597-025-04650-4.

Carfagna, E., Pratesi, M., Carfagna, A., 2013. Methodological developments for improving the reliability and cost-effectiveness of agricultural statistics in developing countries. Hong Kong.

CEOS WGCV, 2025. Focus area on land cover product validation. https://lpvs.gsfc.nasa.gov/LandCover/LC_home.html. CEOS Land Product Validation Subgroup, NASA's Goddard Space Flight Center. Last updated July 17, 2025.

Cochran, W.G., 2002-01-03. Sampling Techniques, third ed. John Wiley.

d'Andrimont, R., Yordanov, M., Martinez-Sanchez, L., Eiselt, B., Palmieri, A., Dominici, P., Gallego, J., Reuter, H.I., Joebges, C., Lemoine, G., van der Velde, M., 2020. Harmonised LUCAS in-situ land cover and use database for field surveys from 2006 to 2018 in the European Union. Sci. Data 7 (1), 352. http://dx.doi.org/10.1038/s41597-020-00675-z, URL: https://www.nature.com/articles/s41597-020-00675-z.

Djenontin, I.N.S., Zulu, L.C., Richardson, R.B., 2022. Smallholder farmers and forest landscape restoration in sub-Saharan Africa: Evidence from Central Malawi. Land Use Policy 122, 106345. http://dx.doi.org/10.1016/j.landusepol.2022.106345, URL: https://www.sciencedirect.com/science/article/pii/S0264837722003726.

Esri and Impact Observatory, 2024. Sentinel-2 10m land use/land cover. URL: https://livingatlas.arcgis.com/landcoverexplorer/. Accessed via Sentinel-2 10m LULC time series (2017–2024). ArcGIS Living Atlas of the World.

FAO, 2023. FAO, 2023 FAOSTAT land, inputs and sustainability, land use. URL: http://www.fao.org/faostat/en/#data/RL.

FAO, 2025. Strengthening trust in food and agricultural statistics: a closer look at the FAO Statistics and Data Quality Assurance Framework. Technical Report, Food and Agriculture Organization of the United Nations, Rome, Italy, URL: https://www.fao.org/statistics/highlights-archive/highlights-detail/strengthening-trust-in-food-and-agricultural-statistics–a-closer-look-at-the-fao-statistics-and-data-quality-assurance-framework/en.

Finegold, Y., Ortmann, A., 2016. Map accuracy assessment and area estimation: a practical guide. FAO, URL: https://openknowledge.fao.org/handle/20.500.14283/i5601e.

Franch, B., Becker-Reshef, I., Cintas, J., Sanchez-Torres, M.J., Roger, J., Skakun, S., Sobrino, J.A., Van Tricht, K., Degerickx, J., Gilliams, S., Koetz, B., Szantoi, Z., Whitcraft, A., 2022. Global crop calendars of maize and wheat in the framework of the WorldCereal project. GISci. Remote. Sens. 59 (1), 885–913. http://dx.doi.org/10.1080/15481603.2022.2079273.

Fritz, S., See, L., McCallum, I., You, L., Bun, A., Moltchanova, E., Duerauer, M., Albrecht, F., Schill, C., Perger, C., Havlik, P., Mosnier, A., Thornton, P., Wood-Sichra, U., Herrero, M., Becker-Reshef, I., Justice, C., Hansen, M., Gong, P., Abdel Aziz, S., Cipriani, A., Cumani, R., Cecchi, G., Conchedda, G., Ferreira, S., Gomez, A., Haffani, M., Kayitakire, F., Malanding, J., Mueller, R., Newby, T., Nonguierma, A., Olusegun, A., Ortner, S., Rajak, D.R., Rocha, J., Schepaschenko, D., Schepaschenko, M., Terekhov, A., Tiangwa, A., Vancutsem, C., Vintrou, E., Wenbin, W., van der Velde, M., Dunwoody, A., Kraxner, F., Obersteiner, M., 2015. Mapping global cropland and field size. Global Change Biol. 21 (5), 1980–1992. http://dx.doi.org/10.1111/gcb.12838, URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.12838. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcb.12838.

Galbally, I.E., 2000. Good practice guidance and uncertainty management in national greenhouse gas inventories: recent developments.

Gallego, F.J., 2012. The efficiency of sampling very high resolution images for area estimation in the European union. Int. J. Remote Sens. 33 (6), 1868–1880. http://dx.doi.org/10.1080/01431161.2011.602993.

Gallego, F.J., Kussul, N., Skakun, S., Kravchenko, O., Shelestov, A., Kussul, O., 2014. Efficiency assessment of using satellite data for crop area estimation in Ukraine. Int. J. Appl. Earth Obs. Geoinf. 29, 22–30.

Giller, K.E., Delaune, T., Silva, J.V., van Wijk, M., Hammond, J., Descheemaeker, K., van de Ven, G., Schut, A.G.T., Taulya, G., Chikowo, R., Andersson, J.A., 2021. Small farms and development in sub-Saharan Africa: Farming for food, for income or for lack of better options? Food Secur. 13 (6), 1431–1454. http://dx.doi.org/10.1007/s12571-021-01209-0.

He, H., Garcia, E.A., 2009. Learning from imbalanced data. IEEE Trans. Knowl. Data Eng. 21 (9), 1263–1284. http://dx.doi.org/10.1109/TKDE.2008.239.

Hu, Q., Yin, H., Friedl, M.A., You, L., Li, Z., Tang, H., Wu, W., 2021. Integrating coarse-resolution images and agricultural statistics to generate sub-pixel crop type maps and reconciled area estimates. Remote Sens. Environ. 258, 112365.

Kebede, E.A., Abou Ali, H., Clavelle, T., Froehlich, H.E., Gephart, J.A., Hartman, S., Herrero, M., Kerner, H., Mehta, P., Nakalembe, C., et al., 2024. Assessing and addressing the global state of food production data scarcity. Nat. Rev. Earth Environ. 5 (4), 295–311.

Kerner, H., Nakalembe, C., Yang, A., Zvonkov, I., McWeeny, R., Tseng, G., Becker-Reshef, I., 2024. How accurate are existing land cover maps for agriculture in Sub-Saharan Africa? Sci. Data 11 (1), 486. http://dx.doi.org/10.1038/s41597-024-03306-z, URL: https://www.nature.com/articles/s41597-024-03306-z. Publisher: Nature Publishing Group.

Kerner, H.R., Nakalembe, C., Yeh, B., Zvonkov, I., Skakun, S., Becker-Reshef, I., McNally, A., 2024. Satellite data shows resilience of tigrayan farmers in crop cultivation during civil war. Sci. Remote. Sens. 10, 100140. http://dx.doi.org/10.1016/j.srs.2024.100140, URL: https://www.sciencedirect.com/science/article/pii/S2666017224000245.

Kerner, H., Tseng, G., Becker-Reshef, I., Nakalembe, C., Barker, B., Munshell, B., Paliyam, M., Hosseini, M., 2020. Rapid response crop maps in Data Sparse Regions. CoRR abs/2006.16866. arXiv:2006.16866. URL: https://arxiv.org/abs/2006.16866.

Liu, X., Yu, L., Li, W., Peng, D., Zhong, L., Li, L., Xin, Q., Lu, H., Yu, C., Gong, P., 2018. Comparison of country-level cropland areas between ESA-CCI land cover maps and FAOSTAT data. Int. J. Remote Sens. 39 (20), 6631–6645.

Lobell, D.B., Azzari, G., Burke, M., Gourlay, S., Jin, Z., Kilic, T., Murray, S., 2020. Eyes in the sky, boots on the ground: Assessing satellite- and ground-based approaches to crop yield measurement and analysis. Am. J. Agric. Econ. 102 (1), 202–219. http://dx.doi.org/10.1093/ajae/aaz051, URL: https://onlinelibrary.wiley.com/doi/abs/10.1093/ajae/aaz051. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1093/ajae/aaz051.

McRoberts, R.E., 2011. Satellite image-based maps: Scientific inference or pretty pictures? Remote Sens. Environ. 115 (2), 715–724. http://dx.doi.org/10.1016/j.rse.2010.10.013, URL: https://linkinghub.elsevier.com/retrieve/pii/S0034425710003184.

McRoberts, R.E., Tomppo, E.O., 2007. Remote sensing support for national forest inventories. Remote Sens. Environ. 110 (4), 412–419. http://dx.doi.org/10.1016/j.rse.2006.09.034, URL: https://www.sciencedirect.com/science/article/pii/S0034425707001721. ForestSAT Special Issue.

Ministry of Agriculture & Livestock Development, 2019. Agricultural census, 2019. https://statistics.kilimo.go.ke/en/1_7/. (Last accessed 21 April 2025).

Mungai, L.M., Messina, J.P., Snapp, S., 2020. Spatial pattern of agricultural productivity trends in Malawi. Sustainability 12 (4), http://dx.doi.org/10.3390/su12041313, URL: https://www.mdpi.com/2071-1050/12/4/1313.

Nakalembe, C., Becker-Reshef, I., Bonifacio, R., Hu, G., Humber, M.L., Justice, C.J., Keniston, J., Mwangi, K., Rembold, F., Shukla, S., et al., 2021. A review of satellite-based global agricultural monitoring systems available for Africa. Glob. Food Secur. 29, 100543.

National Bureau of Statistics, Tanzania, 2021. National sample census of agriculture 2019/20. https://www.nbs.go.tz/uploads/statistics/documents/sw-1705482872-2019-20_Agri_Census_Atlas.pdf. (Last accessed 21 April 2025).

National Institute of Statistics of Rwanda, 2019. Rwanda seasonal agricultural survey - 2019 annual report. https://www.statistics.gov.rw/publication/seasonal-agricultural-survey-2019-annual-report. (Last accessed 21 April 2025).

Neigh, C.S.R., Masek, J.G., Nickeson, J.E., 2013. High-resolution satellite data open for government research. Eos, Trans. Am. Geophys. Union 94 (13), 121–123. http://dx.doi.org/10.1002/2013EO130002, URL: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2013EO130002.

Nzabarinda, V., Bao, A., Xu, W., Uwamahoro, S., Huang, X., Gao, Z., Umugwaneza, A., Kayumba, P.M., Maniraho, A.P., Jiang, Z., 2021. Impact of cropland development intensity and expansion on natural vegetation in different African countries. Ecol. Inform. 64, 101359. http://dx.doi.org/10.1016/j.ecoinf.2021.101359, URL: https://www.sciencedirect.com/science/article/pii/S1574954121001503.

Oliphant, A., Thenkabail, P., Teluguntla, P., 2022. Global Food-Security-Support-Analysis Data at 30-M Resolution (GFSAD30) Cropland-Extent Products—Download Analysis. Technical Report, US Geological Survey.

Olofsson, P., Arévalo, P., Espejo, A.B., Green, C., Lindquist, E., McRoberts, R.E., Sanz, M.J., 2020. Mitigating the effects of omission errors on area and area change estimates. Remote Sens. Environ. 236, 111492. http://dx.doi.org/10.1016/j.rse.2019.111492, URL: https://www.sciencedirect.com/science/article/pii/S0034425719305115.

Olofsson, P., Foody, G.M., Herold, M., Stehman, S.V., Woodcock, C.E., Wulder, M.A., 2014. Good practices for estimating area and assessing accuracy of land change. Remote Sens. Environ. 148, 42–57. http://dx.doi.org/10.1016/j.rse.2014.02.015, URL: https://www.sciencedirect.com/science/article/pii/S0034425714000704.

Palmer, P.I., Wainwright, C.M., Dong, B., Maidment, R.I., Wheeler, K.G., Gedney, N., Hickman, J.E., Madani, N., Folwell, S.S., Abdo, G., Allan, R.P., Black, E.C.L., Feng, L., Gudoshava, M., Haines, K., Huntingford, C., Kilavi, M., Lunt, M.F., Shaaban, A., Turner, A.G., 2023. Drivers and impacts of Eastern African rainfall variability. Nat. Rev. Earth Environ. 4 (4), 254–270. http://dx.doi.org/10.1038/s43017-023-00397-x, URL: https://www.nature.com/articles/s43017-023-00397-x. Publisher: Nature Publishing Group.

Potapov, P., Turubanova, S., Hansen, M.C., Tyukavina, A., Zalles, V., Khan, A., Song, X.-P., Pickens, A., Shen, Q., Cortez, J., 2022. Global maps of cropland extent and change show accelerated cropland expansion in the twenty-first century. Nat. Food 3 (1), 19–28. http://dx.doi.org/10.1038/s43016-021-00429-z, URL: https://www.nature.com/articles/s43016-021-00429-z. Publisher: Nature Publishing Group.

Skakun, S., 2025. The impact of map accuracy on area estimation with remotely sensed data within the stratified random sampling design. Remote Sens. Environ. 326, 114805. http://dx.doi.org/10.1016/j.rse.2025.114805, URL: https://linkinghub.elsevier.com/retrieve/pii/S0034425725002093.

Stehman, S.V., 2014. Estimating area and map accuracy for stratified random sampling when the strata are different from the map classes. Int. J. Remote Sens. 35 (13), 4923–4939. http://dx.doi.org/10.1080/01431161.2014.930207, Publisher: Taylor & Francis.

Stehman, S.V., Foody, G.M., 2019. Key issues in rigorous accuracy assessment of land cover products. Remote Sens. Environ. 231, 111199. http://dx.doi.org/10.1016/j.rse.2019.05.018, URL: https://www.sciencedirect.com/science/article/pii/S0034425719302111.

Stehman, S.V., Mousoupetros, J., McRoberts, R.E., Næsset, E., Pengra, B.W., Xing, D., Horton, J.A., 2022. Incorporating interpreter variability into estimation of the total variance of land cover area estimates under simple random sampling. Remote Sens. Environ. 269, 112806. http://dx.doi.org/10.1016/j.rse.2021.112806, URL: https://www.sciencedirect.com/science/article/pii/S0034425721005265.

Stehman, S.V., Wagner, J.E., 2024. Choosing a sample size allocation to strata based on trade-offs in precision when estimating accuracy and area of a rare class from a stratified sample. Remote Sens. Environ. 300, 113881. http://dx.doi.org/10.1016/j.rse.2023.113881, URL: https://linkinghub.elsevier.com/retrieve/pii/S0034425723004327.

Stehman, S.V., Wickham, J.D., 2011. Pixels, blocks of pixels, and polygons: Choosing a spatial unit for thematic accuracy assessment. Remote Sens. Environ. 115 (12), 3044–3055. http://dx.doi.org/10.1016/j.rse.2011.06.007, URL: https://www.sciencedirect.com/science/article/pii/S0034425711002318.

Tsendbazar, N., Herold, M., Li, L., Tarko, A., De Bruin, S., Masiliunas, D., Lesiv, M., Fritz, S., Buchhorn, M., Smets, B., Van De Kerchove, R., Duerauer, M., 2021. Towards operational validation of annual global land cover maps. Remote Sens. Environ. 266, 112686. http://dx.doi.org/10.1016/j.rse.2021.112686, URL: https://linkinghub.elsevier.com/retrieve/pii/S0034425721004065.

Tseng, G., Kerner, H., Nakalembe, C., Becker-Reshef, I., 2020. Annual and in-season mapping of cropland at field scale with sparse labels. In: Proceedings of the Thirty-Fourth Conference on Neural Information Processing Systems Workshops. URL: https://s3.us-east-1.amazonaws.com/climate-change-ai/papers/neurips2020/29/paper.pdf.

Tubiello, F.N., Conchedda, G., Casse, L., Hao, P., De Santis, G., Chen, Z., 2023a. A new cropland area database by country circa 2020. Earth Syst. Sci. Data 15 (11), 4997–5015. http://dx.doi.org/10.5194/essd-15-4997-2023, URL: https://essd.copernicus.org/articles/15/4997/2023/. Publisher: Copernicus GmbH.

Tubiello, F.N., Conchedda, G., Casse, L., Pengyu, H., Zhongxin, C., De Santis, G., Fritz, S., Muchoney, D., 2023b. Measuring the world's cropland area. Nat. Food 4 (1), 30–32. http://dx.doi.org/10.1038/s43016-022-00667-9, URL: https://www.nature.com/articles/s43016-022-00667-9.

Tubiello, F.N., Conchedda, G., de Santis, G., Wanner, N., 2021. Land Use Statistics and Indicators Statistics. Global, Regional and Country Trends 1990–2019. Technical Report, FAO, Rome, p. 14.

Tyukavina, A., Stehman, S.V., Pickens, A.H., Potapov, P., Hansen, M.C., 2025. Practical global sampling methods for estimating area and map accuracy of land cover and change. Remote Sens. Environ. 324, 114714. http://dx.doi.org/10.1016/j.rse.2025.114714, URL: https://www.sciencedirect.com/science/article/pii/S003442572500118X.

Uganda Bureau of Statistics, 2020. Annual agricultural survey 2019 report. https://www.ubos.org/wp-content/uploads/publications/04_2022AAS2019_Report.pdf. (Last accessed 21 April 2025).

Van Tricht, K., Degerickx, J., Gilliams, S., Zanaga, D., Battude, M., Grosu, A., Brombacher, J., Lesiv, M., Bayas, J.C.L., Karanam, S., Fritz, S., Becker-Reshef, I., Franch, B., Mollà-Bononad, B., Boogaard, H., Pratihast, A.K., Koetz, B., Szantoi, Z., 2023. WorldCereal: a dynamic open-source system for global-scale, seasonal, and reproducible crop and irrigation mapping. Earth Syst. Sci. Data 15 (12), 5491–5515. http://dx.doi.org/10.5194/essd-15-5491-2023, URL: https://essd.copernicus.org/articles/15/5491/2023/. Publisher: Copernicus GmbH.

Waldner, F., Jacques, D.C., Löw, F., 2017. The impact of training class proportions on binary cropland classification. Remote. Sens. Lett. 8 (12), 1122–1131. http://dx.doi.org/10.1080/2150704X.2017.1362124.

Whitcraft, A.K., Vermote, E.F., Becker-Reshef, I., Justice, C.O., 2015. Cloud cover throughout the agricultural growing season: Impacts on passive optical earth observations. Remote Sens. Environ. 156, 438–447.

Zambia Statistics Agency, 2020. Crop forecast survey. https://www.zamstats.gov.zm/zamstats/wp-content/uploads/2024/01/CFS-2019_2020.xlsx. (Last accessed 21 April 2025).

Zhang, X., Zhao, T., Xu, H., Liu, W., Wang, J., Chen, X., Liu, L., 2024. GLC_FCS30D: the first global 30 m land-cover dynamics monitoring product with a fine classification system for the period from 1985 to 2022 generated using dense-time-series landsat imagery and the continuous change-detection method. Earth Syst. Sci. Data 16 (3), 1353–1381. http://dx.doi.org/10.5194/essd-16-1353-2024, URL: https://essd.copernicus.org/articles/16/1353/2024/.